Transcribing Western modern manuscripts (1500-2020): an economical, ecological and secured approach

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Research in the Humanities requires large datasets that often exist only in analogue format. Extracting textual information from manuscripts and printed books is therefore a crucial step for the future of computational humanities, as has been shown by recent works in history (Gordijn 2023), historical linguistics (Gabay and Clérice 2024), etc.

Since OCRopy (Breuel 2014), several Optical Character Recognition (OCR) engines have been developed to extract the text from digital facsimiles, such as PyLaia (Puigcerver 2017; Tarride et al. 2024), Kraken (Kiessling 2019) or Calamari (Wick, Reul, and Puppe 2020) – to name just a few. Several all-in-one platforms have been built around these OCR engines, simplifying the user experience to annotate and train models for layout analysis, OCR, reading order, etc. such as Transkribus (Colutto et al. 2019; Kahle et al. 2017), eScriptorium (Kiessling et al. 2019; Stokes et al. 2021) or OCR4all (Reul et al. 2019).

One of the main problems with OCR engines is the cost of the infrastructure required for training models, which leaves two possibilities. On the one hand, one can become a paid (and therefore closed) solution to continue providing the necessary computing power; on the other hand, one can remain a completely open software, but leave it up to researchers to buy and manage their own hardware.

Both options have their respective merits, but we want to draw attention here to two positive consequences of an open solution, despite the technical complexities it can potentially generate. First, it remains possible to deploy a lightweight version (i.e. without a training option) of open tools on any machine for institutions and researchers who cannot afford a paid one (Salar Khan, Umer, and Faruqe 2024). Second, an open-source solution allows IT security teams to control the code and run it on premises in a completely sealed environment – something that is absolutely necessary for highly sensitive data like legal or medical documents (Herzog, Celik, and Sulaiman 2024).

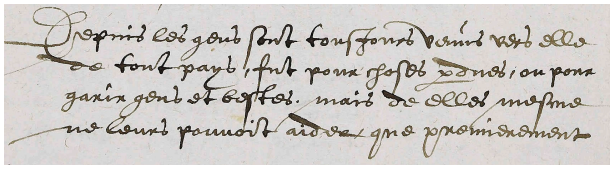
In this paper, we evaluate the possibility of creating a diachronic multilingual model for Western European languages. This massive model is produced under complex conditions, by aggregating data from various projects that have different transcription guidelines. Despite this, it allows for producing reasonable results from scratch, or to be fine-tuned on CPUs only, in case researchers do not have access to GPUs for financial, security, or ecological reasons (Pendergrass et al. 2019).

Massive models, complex data

We know that multilingual models offer promising results (Capurro, Provatorova, and Kanoulas 2023). Transkribus has released in 2023 the “Titan supermodel” (Trankribus

2023), a multilingual transformer-based model whose efficiency is proven to be excellent with complicated historical documents such as postcards in different languages (Smits et al. 2023), even if it now seems challenged by multimodal Large Language Models (LLMs, Humphries et al. 2024). LLMs are based on a similar approach (massive and multilingual models), but because they were (still) paid solutions and did not (yet) work in a sealed environment when we began this study, they were excluded.

Open alternatives have recently emerged, notably for medieval manuscripts (Clérice et al. 2024; Pinche et al. 2024), or prints with Gothic (Solfrini, Gabay, Humeau, et al. 2024) or Latin (Gabay, Clérice, et al. 2024) characters. In addition to being open, these models also have the particularity of being based on homogeneous and documented transcription guidelines, prepared by experts in philology (e.g. Pinche 2022; Solfrini, Gabay, Gross, et al. 2023), coordinated by the CATMuS project (Pinche et al. 2024).

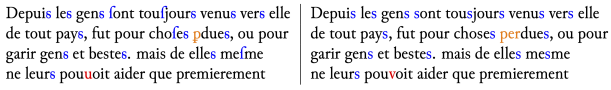


A close up of a handwritten letter AI-generated content may be incorrect.

Figure 1: Archives de l’ancien Évêché de Bâle, Porrentruy, Switzerland, B 168/14-17.1 (1589), f° 1 r.

Initial tests have been conducted for French modern manuscripts with encouraging, but not yet satisfactory, results, whether it is for the Manu McFrench model (Chagué, Clérice, et al. 2023) or the Manu McFondue model (Gabay, Pinche, et al. 2024). One of the main problems, on top of the extreme diversity of the hands, is the existence of many datasets prepared without coordinated transcription guidelines. The development of abbreviations, the transcription of allographs, etc. is therefore different from one project to another

(cf. fig. 1 and 2), which considerably complicates the training task.



A close up of a sign AI-generated content may be incorrect.

Figure 2: Two possible transcriptions of fig. 1, with an abbreviation in orange, variations of *sin* blue, and a ramist letter in red.

Experiment

Data. Gathering data is not trivial: although many projects have their datasets referenced on HTR-United (Chagué and Clérice 2023), it is not the case for all of them (copyright issues, lack of knowledge of the project, etc.). All data used for our experiment is listed in the tab. 1, the bulk of which is written in Dutch, German and French – the datasets for Latin and Spanish are of much smaller magnitude.

As for diversity, the small number of projects in Dutch hides a great diversity of documents, carefully selected by specialists. The data in German are in large quantity, but less diversified in their graphic form, while those in French are more diversified, but to the detriment of the homogeneity of transcription standards.

Project	Language	Century	Pages	Lines	available
IJSBERG	nl	17th-18th c.	6 275	259 517	Yes
REPUBLIC	nl	17th c.	514	46 690	Yes
Subtotal			6 789	306 207	
Heinrich Wölfflin	de/it	20th c.	1 107	2 290	Yes
EditionEugen-Huber	de	20th c.	1 921	45 194	No
Swiss Federal Council	de	19th c.	2 752	2 752	Yes
Staatsarchivs Zürich	de	19th c.	2 969	94 201	No
Meyer von Schauensee	de	18th c.	242	9 303	Yes
Crimes et châtiments	de	17th c.	95	1 756	Yes
Basler Urfehdebuchs X	de	16th c.	192	8 221	Yes
Crimes et châtiments	de	16th c.	65	1 269	Yes
Subtotal			9 343	164 986	
Ricardo Viñes	sp	20th c.	48	2 168	Soon
Occupation of Araucania	sp	19th c.	146	3 235	Partially
Subtotal			194	5 403	
Philiumm	la	17th c.	207	13 049	Partially
Sardi's Toponomasia	la	16th c.	51	2 003	Partially
Subtotal			258	15 052	
Moonshines	fr	21th c.	54	1 186	
CREMMAwikipedia	fr	21th c.	357	1 973	Yes
CREMMAMSS-20	fr	20th c.	15	224	Yes
Digital Peraire	fr	20th c.	75	2 569	Yes

Testaments de Poilus	fr	20th c.	228	3 330	Yes
Verbier Time Machine	fr	19th-20th c.	192	28 476	Partially
Lectaurep (répertoires)	fr	19th-20th c.	223	29 411	Yes
FoNDUE-FR-MSS-19	fr	19th c.	67	1 540	Yes
Timeus	fr	19th c.	250	7 701	Yes
Lectaurep (Mariages)	fr	19th c.	107	20 304	Yes
Private archives	fr	19th c.	82	2 235	No
Proto4DigEd	fr	19th c.	107	1 400	No
CREMMA-MSS-19	fr	19th c.	79	1 807	Yes
FoNDUE-FR-MSS-18	fr	18th c.	333	8 496	Yes
Masculinités esclavagistes	fr	18th c.	282	7 566	Yes
CREMMA-MSS-18	fr	18th c.	136	4 019	Yes
Lectaurep (Bronod)	fr	18th c.	101	3 707	Yes
FoNDUE-FR-MSS-17	fr	17th c.	8	91	Yes
CREMMA-MSS-17	fr	17th c.	115	2 213	Yes
Crimes et châtiments	fr	17th c.	882	22 359	Yes
HTRomance	fr	16th-19th c.	173	3 385	Yes
Crimes et châtiments	fr	16th c.	500	14 856	Yes
Subtotal			4 177	168 848	
Total			20 761	660 496	

Table 1: Data used for training.

Preparation. When needed, the data in PageXML are transformed into an ALTO format. For documents from Transkribus, the height of the baseline is controlled and corrected if it is too low (a few pixels less vertically). The masks are all repolygonised with Kraken 5.2, and the characters are harmonised with Chocomufin (e.g. straight vs curly quotes).

Training. The model is trained with Kraken. We use a batch of 70, a learning rate of $\approx\sqrt{70}\cdot 10^{-4}$ (i.e. 8e−4), following Kraken's recommendations, an automatic mixed precision, a NFD (Normalization Form canonical Decomposition) Unicode normalisation and data augmentation. Rather than XML files, we use precompiled binary dataset files. The model produced is called FoNDUE-GD and its training lasted 30 epochs.

In a second step, we produce a model per linguistic domain by fine-tuning FoNDUE-GD only on data from the same language, still with the same configuration, except the batch size (16) and the learning rate (4e−4).

Test set	Language	Century	ManuMcFondue	FoNDUE-GD	FoNDUE-GD-ft
Test 1 (random selection)	multi	16th-21th c.	73.54	89.05	
Test 2 (curated selection)	multi	16th-21th c.	76.43	85.30	
REPUBLIC	nl	17th c.	85.00	96.36	97.13
IJSBERG	nl	18th c.	89.09	93.13	95.01
IJSBERG	nl	19th c.	94.60	93.60	95.04
IJSBERG	nl	20th c.	93.41	91.41	94.17
EditionEugenHuber	de	20th c.	44.81	90.72	93.20
Staatsarchiv Zurich 1	de	19th c.	66.02	92.08	93.31
Staatsarchiv Zurich 2	de	19th c.	54.24	94.14	94.97
Meyer von Schauensee	de	18th c.	70.35	87.45	88.31
Crimes et châtiments	de	16th-17th c.	55.81	75.22	80.17
Basler Urfehdebuchs X	de	16th c.	78.98	22.09	43.27
Ricardo Viñes	sp	20th c.	52.00	74.53	84.57
Occupation of Araucania	sp	19th	80.87	82.59	88.69
Philiumm	la	17th c.	68.46	90.78	94.86
Sardi's Toponomasia	la	16th c.	88.14	88.27	92.86
Moonshines	fr	21th c.	67.84	84.38	89.26
CREMMA wikipedia	fr	21th c.	69.09	79.47	83.42
Testaments de Poilus	fr	20th c.	71.77	77.56	81.13
Digital Peraire	fr	20th c.	71.12	82.32	86.63
Verbier Time Machine	fr	19th-20th c.	85.44	83.68	87.82
Lectaurep (répertoires) 1 -Dufour	fr	19th-20th c.	92.03	86.73	90.52
Lectaurep (répertoires) 2	fr	19th-20th c.	86.63	83.19	86.51
FoNDUE-FR-MSS-19 (Zola)	fr	19th c.	78.24	79.32	84.34
Timeus	fr	19th c.	60.52	83.61	87.23
Lectaurep (mariages)	fr	19th-20th c.	89.38	88.64	90.30
Private archives	fr	19th c.	86.88	82.73	87.03
Proto4DigEd	fr	19th c.	51.54	70.71	79.98
FoNDUE-MSS-18 (Bastille)	fr	18th c.	78.47	86.58	90.74
FoNDUE-MSS-18 (R.~Simon)	fr	18th c.	92.77	92.24	93.22
HTRomance	fr	17th-18th c.	80.08	86.71	89.14
Crimes et châtiments	fr	16th-17th c.	63.49	82.98	88.00

Table 2: Accuracy for each model on generic and specific test sets. ManuMcFondue serves as a baseline. FoNDUE-GD-ft are versions of FoNDUE-GD fine-tuned on all the data in the same language that those tested. Best scores are in bold.

In-domain testing

Testing the effectiveness of the model is not trivial, because the corpus is heavily imbalanced (up to 1:35 for the ratio between pages in Latin and in Spanish). To skirt this problem, two test sets are prepared.

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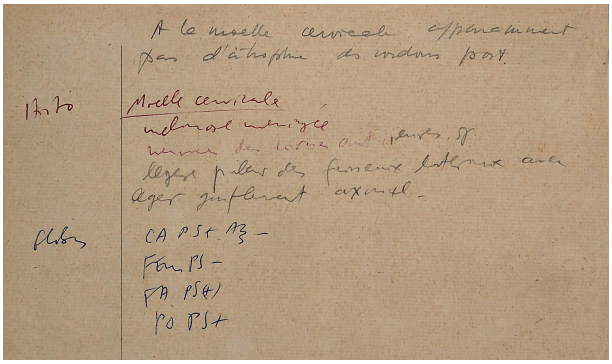
  - The first one is a random selection of 2% of the pages (i.e. c. 420 pages, cf. Test 1 in tab. 2), which offers an evaluation of the model relative to the training data. In practice: there are a lot of data in Dutch, much less in Latin.
  - The second one uses a curated selection of 4 pages taken from different projects. By doing so we increase the representation of scarce classes: there are, relatively to the corpus, more data in Latin, and the ratio between data in Latin and in Dutch is now of 1:2 in the test set. Therefore, we make possible an evaluation on partially rebalanced data (cf. Test 2 in tab. 2), but also the execution of separate tests for each class for a more detailed study of the performance of the model.

The results are extremely satisfactory, with an accuracy of c. 85-90% for almost all languages, and all periods. Some projects that have not seen the amount of data increase between the training of Manu McFondue see their prediction improve only thanks to the addition of new data, and the heterogeneity of the transcription practices seems to have a minimal impact.

The idea of fine-tuning specific models for each linguistic domain from the massive model seems to be a profitable strategy, and significantly improves the quality of the prediction. The languages least well represented in the train set of the massive model are logically those which are experiencing the strongest increase with linguistic fine-tuning, certainly because it counterbalances the small quantity of data used when training the massive model and because the data are less diverse.

Experiment 2



A close-up of a piece of paper AI-generated content may be incorrect.

A la moelle cervicale apparamment
pas d'atrophie des cordons post
Moelle cervicale
melanose meningée nervures des cornes antérieures. SP.
legere paleur des faisseaux latéraux avec
leger gonflement axonal.
CA PS+ Alz-
FEm PS-
FA PS (+)
PO PS +.

Figure 3: Example of a medical document. We observe several hands, several pens, slightly erased areas, etc.

In order to more precisely measure the efficiency of such secondary models, we propose another experiment: we test FoNDUE-GD on out-of-domain (OOD) data. We choose two new datasets:

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  - Italian data, a language (almost) unknown to FoNDUE-GD. This dataset is made of manuscripts from the 18th century written by historian Ludovico Antonio Muratori (1672-1750) (Cascianelli et al. 2022).
  - Medical data from the 20th century, the form and lexical content of which also diverge strongly from those of the massive model (cf. fig. 3).

Fine-tuning is done with only one CPU. We try to fine-tune the generic model, FoNDUE-GD, and its fine-tuned version on French data, FoNDUE-GD-fr, because Italian is also a Romance language and the medical data are in French.

Type	Language	Century	Pages (test)	Lines (Text)	FoNDUE-GD	FoNDUE-GD-ft	FoNDUE-GD-fr-ft
Correspondance	It	18 th	15 (2)	334 (37)	80.68	86.06	88.51
Medical records	Fr	20 th	156 (5)	2427 (41)	68.35	75.93	77.37

Table 3: Out-of-domain test sets with fine-tuning scores. FoNDUE-GD is the baseline, FoNDUE-GD-ft is the fine-tuned version of FoNDUE-GD on the OOD data, FoNDUE-GD-fr-ft is the fine-tuned version of FoNDUE-GD-fr on the OOD data.

Once again, the results are extremely satisfactory, with an accuracy of slightly less than 90% with 13 pages in Italian, and almost 80% for extremely complicated medical data.

Conclusion

FoNDUE-GD should not (only) be seen as a mega model, but as a primary model from which secondary models dedicated to specific tasks are derived. While our model can be used out of the box for many documents, it can also be fine-tuned with limited data to be adapted to specific scenarios, such as prediction in a new language or for complex documents.

Given the impossibility of correcting all the transcriptions contained in our dataset, the question of data harmonization is the next challenge before us. One solution would be to evaluate how quickly our massive model can "unlearn" the conflicting rules it has seen by fine-tuning it with a homogeneous training set.

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